Auction theory in practice
Lecture 2

Esther David

Sequential Move Games

Partially based on:

Mike Shor slides for Game Theory & Business Strategy
Robert Marks, NSW

Review

- Understanding the outcomes of games
- Sometimes easy
  - Dominant strategies
- Sometimes more challenging
  - “I know that you know …”

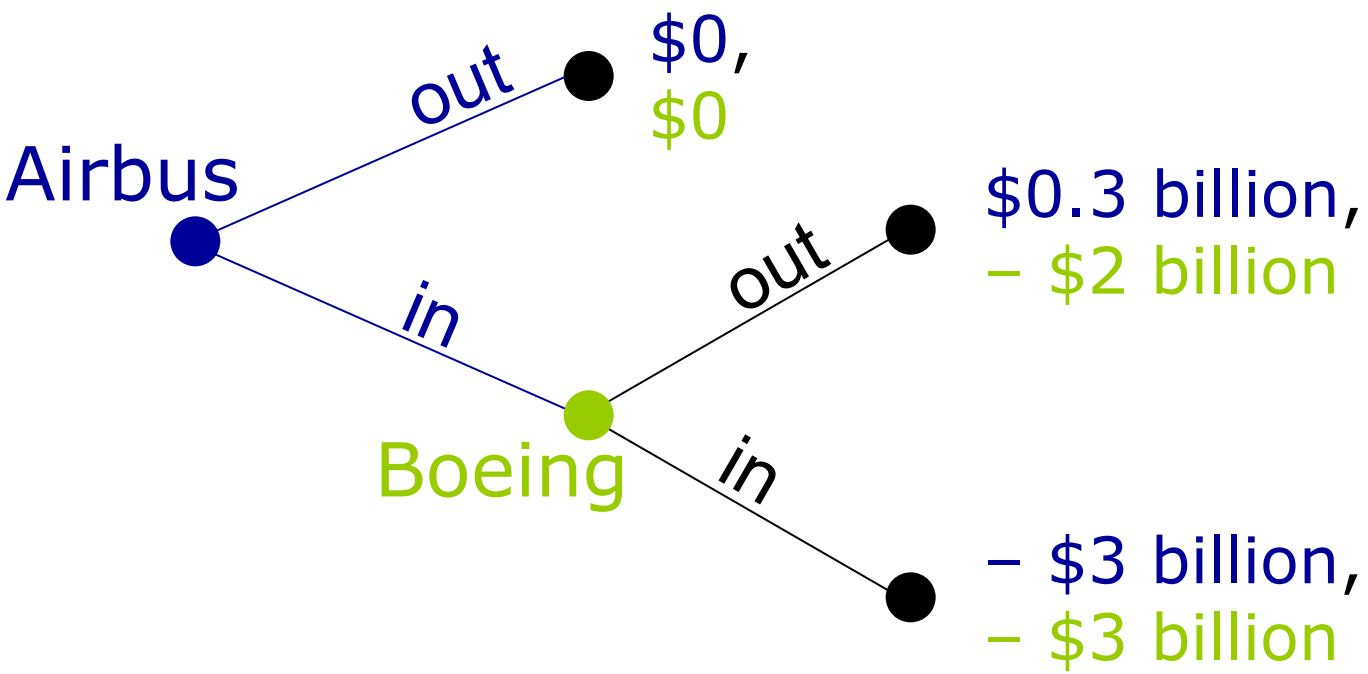
- What if a game is sequential?
  - Market entry

Very Large Airplanes: Airbus vs. Boeing

- Airbus lacked a jumbo jet:
  - " The problem is the monopoly of the 747 … They have a product. We have none. "
    - Airbus Executive
  - Initiated plans to build a super-jumbo jet
- Industry feasibility studies:
  - Room for at most one megaseater
- Boeing preemption:
  - "Boeing, the world's top aircraft maker, announced it was building a plane with 600 to 800 seats, the biggest and most expensive airliner ever."
    - BusinessWeek, 1993

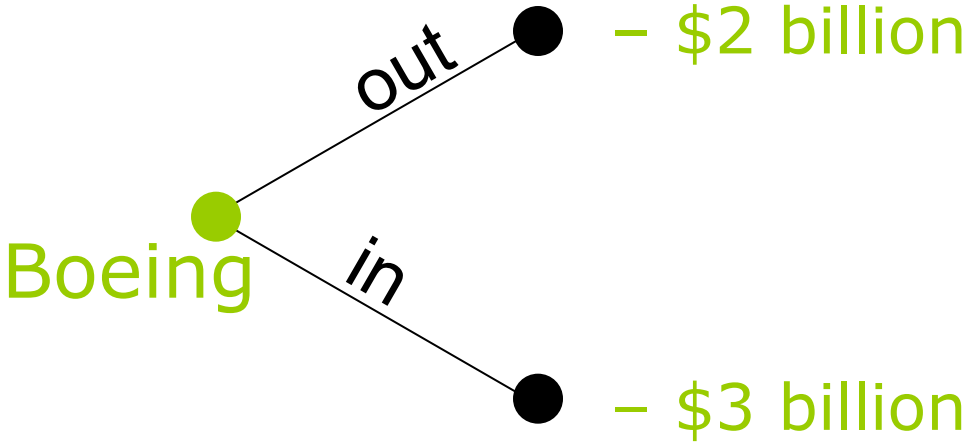
Sequential Games

The Game



Looking Forward …

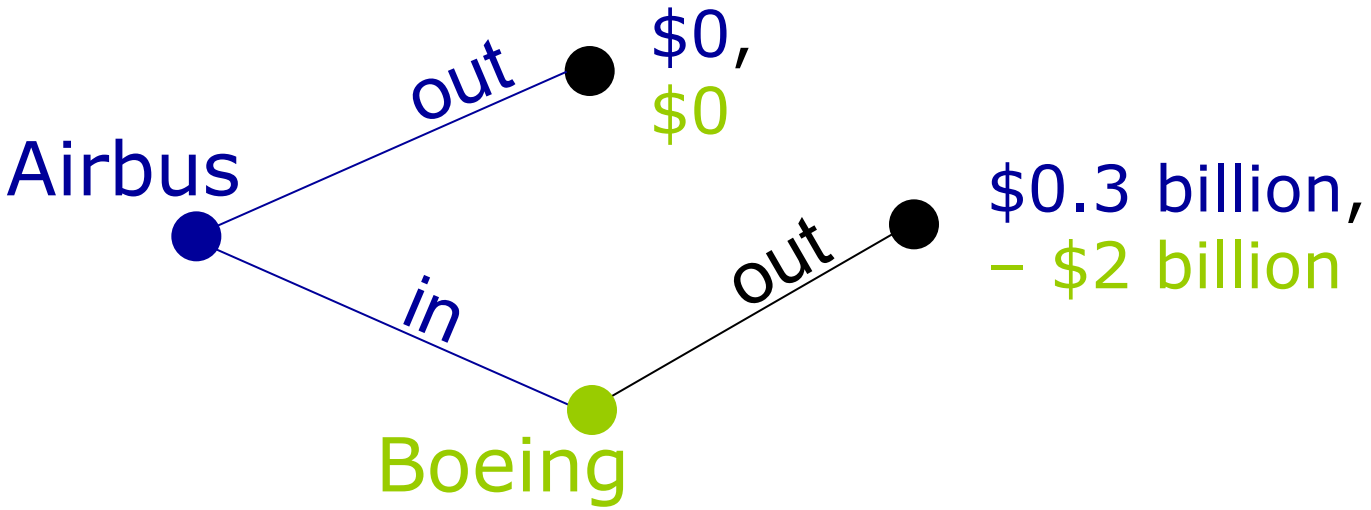
- Airbus makes the first move:
  - Must consider how Boeing will respond
- If enter:



- Boeing accommodates, stays out

… And Reasoning Back

- Now consider the first move:



- Only (In, Out) is *sequentially rational*
  - In is not credible (for Boeing)

Extensive Form Games

aka Dynamic Games, aka Tree-Form Games

- Extensive form games allows us to model situations where agents take actions over time
- Simplest type is the perfect information game

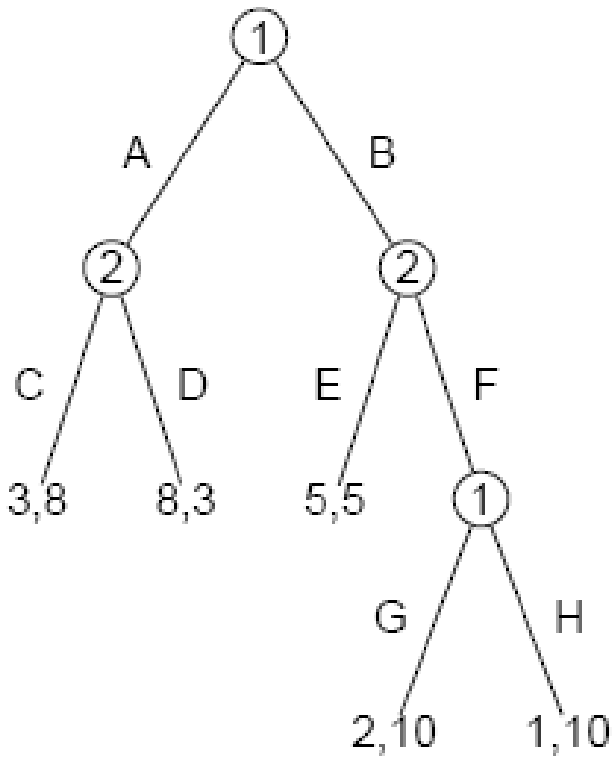
Tree Representation

- The definition is really a tree description
- Each node is defined by its history (sequence of nodes leading from root to it)
- The descendents of a node are all choice and terminal nodes in the subtree rooted at the node.

Strategies

- A strategy, s_i of player i is a function that assigns an action to each non-terminal history, at which the agent can move.
- Outcome: $o(s)$ of strategy profile s is the terminal history that results when agents play s
- **Important:** The strategy definition requires a decision at each choice node, regardless of whether or not it is possible to reach that node given earlier moves

Example



Strategy sets for the agents

$$S_1 = \{(A,G),(A,H),(B,G),(B,H)\}$$

$$S_2 = \{(C,E),(C,F),(D,E),(D,F)\}$$

Example

We can transform an extensive form game into a normal form game.

	(C,E)	(C,F)	(D,E)	(D,F)
(A,G)	3,8	3,,8	8,3	8,3
(A,H)	3,8	3,,8	8,3	8,3
(B,G)	5,5	2,10	5,5	2, 10
(B,H)	5,5	1,0	5,5	1,0

Nash Equilibria

Definition (Nash Equilibrium)

Strategy profile s^* is a Nash Equilibrium in a perfect information, extensive form game if for all i

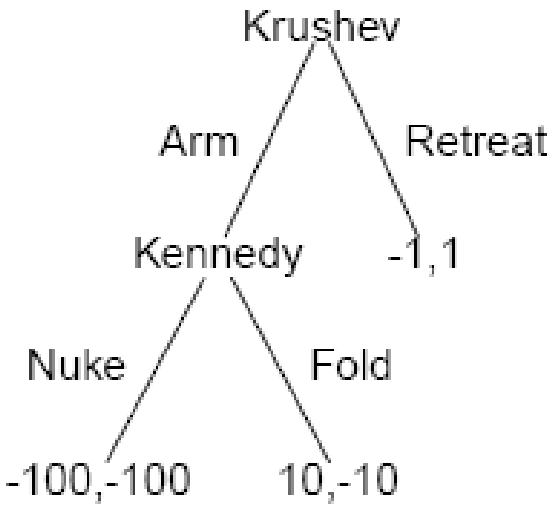
$$u_i(s_i^*, s_{-i}^*) \geq u_i(s_i', s_{-i}^*) \forall s_i'$$

Theorem

Any perfect information game in extensive form has a pure strategy Nash equilibrium.

Intuition: Since players take turns, and everyone sees each move there is no reason to randomize.

Example: Bay of Pigs



What are the NE?

Subgame Perfect Equilibrium

Nash Equilibrium can sometimes be too weak a solution concept.

Definition (Subgame)

Given a game G, the subgame of G rooted at node j is the restriction of G to its descendents of h.

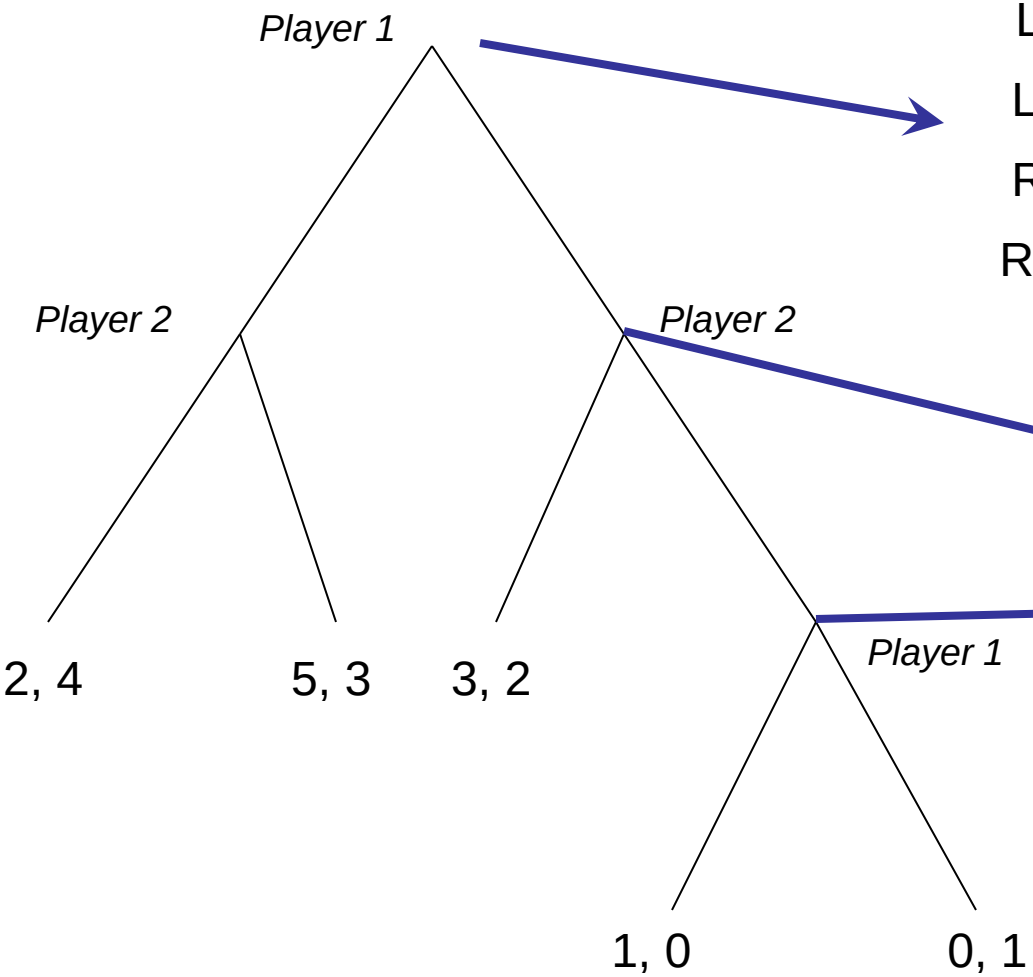
Definition (Subgame perfect equilibrium)

A strategy profile s^* is a subgame perfect equilibrium if for all $i \in N$, and for all subgames of G, the restriction of s^* to G' (G' is a subgame of G) is a Nash equilibrium in G'. That is

$$\forall i, \forall G', u_i(s_i^*|_{G'}, s_{-i}^*|_{G'}) \geq u_i(s_i'|_{G'}, s_{-i}^*|_{G'}) \forall s_i'$$

Subgame perfect equilibrium

- Each node in a (perfect-information) game tree, together with the remainder of the game after that node is reached, is called a subgame
- A strategy profile is a subgame perfect equilibrium if it is an equilibrium for every subgame



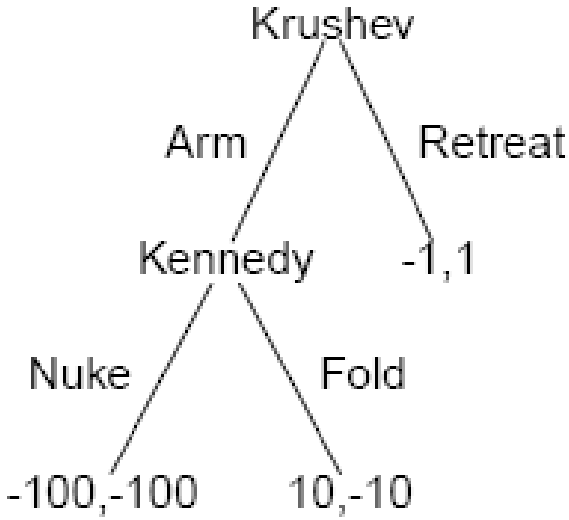
	LL	LR	RL	RR
LL	2, 4	2, 4	5, 3	5, 3
LR	2, 4	2, 4	5, 3	5, 3
RL	3, 2	1, 0	3, 2	1, 0
RR	3, 2	0, 1	3, 2	0, 1

	*L	*R
*L	3, 2	1, 0
*R	3, 2	0, 1

	**
*L	1, 0
*R	0, 1

- (RR, LL) and (LR, LR) are not subgame perfect equilibria because (*R, **) is not an equilibrium
- (LL, LR) is not subgame perfect because (*L, *R) is not an equilibrium
  - *R is not a credible threat

Example: Bay of Pigs



What are the SPE?

Existence of SPE

Theorem (Kuhn's Thm)

Every finite extensive form game with perfect information has a SPE.

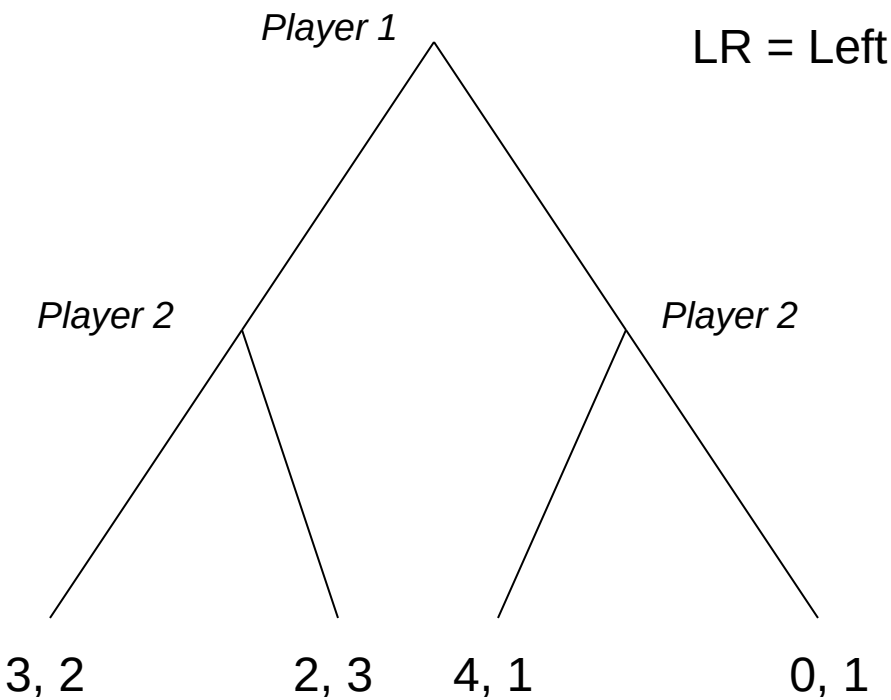
You can find the SPE by backward induction.

- Identify equilibria in the bottom-most trees
- Work upwards

Rollback, or Backwards Induction

1. From the terminal nodes (final payoffs), go up the tree to the first parent decision nodes.
2. Identify the best decision for the deciding player at each node.
3. "Prune" all branches from the decision node in 2. Put payoffs at new end = best decision's payoffs
4. Do higher decision nodes remain? If "no", then finish.
5. If "yes", then go to step 1.
6. For each player, the collection of best decisions at each decision node of that player $\rightarrow$ best strategies of that player.

Conversion from extensive to normal form

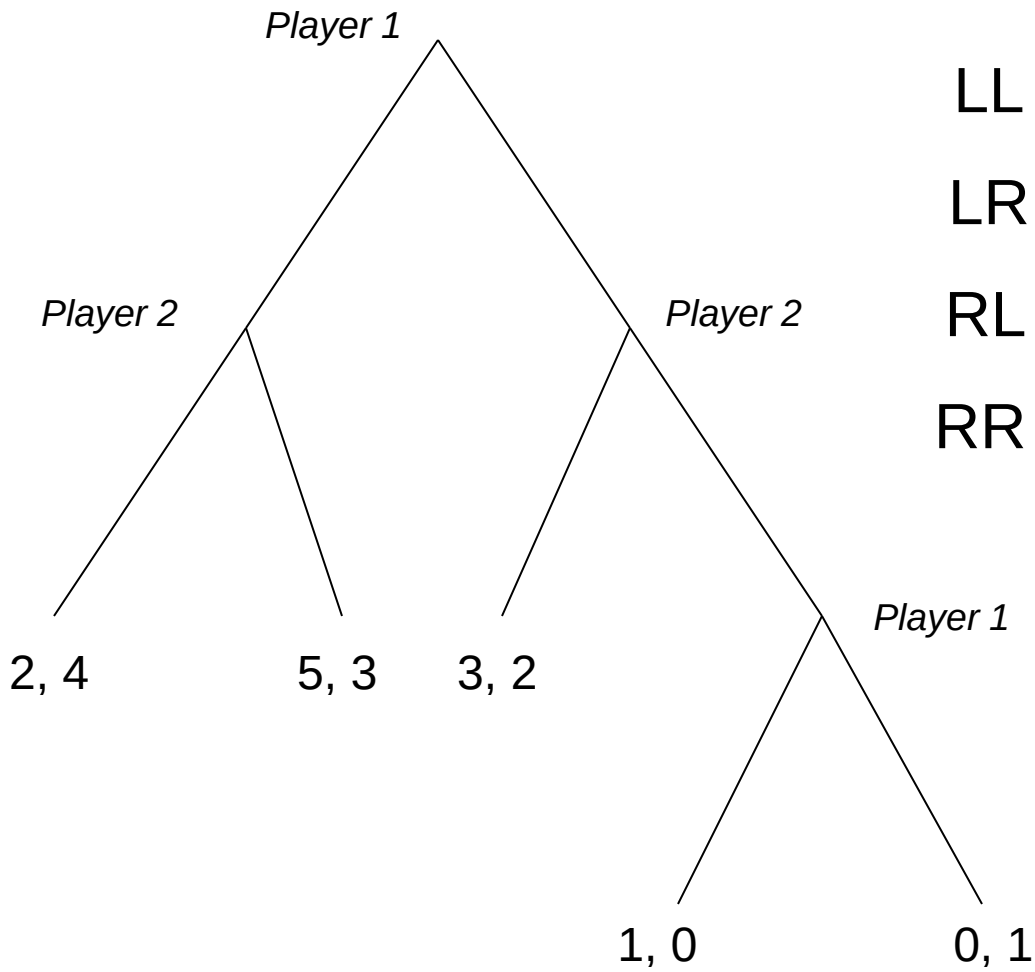


LR = Left if 1 moves Left, Right if 1 moves Right; etc.

	LL	LR	RL	RR
L	3, 2	3, 2	2, 3	2, 3
R	4, 1	0, 1	4, 1	0, 1

- Nash equilibria of this normal-form game include (R, LL), (R, RL), (L, RR) + infinitely many mixed-strategy equilibria
- In general, normal form can have exponentially many strategies

Converting the first game to normal form



	LL	LR	RL	RR
LL	2, 4	2, 4	5, 3	5, 3
LR	2, 4	2, 4	5, 3	5, 3
RL	3, 2	1, 0	3, 2	1, 0
RR	3, 2	0, 1	3, 2	0, 1

- Pure-strategy Nash equilibria of this game are (LL, LR), (LR, LR), (RL, LL), (RR, LL)
- But the only backward induction solution is (RL, LL)

The Capacity Game Revisited

Last lecture the two firms Alpha and Beta simultaneously made the capacity decision:

		Beta	
	DNE	Small	Large
DNE	\$18, \$18	\$15, \$20	\$9, \$18
Alpha Small	\$20, \$15	\$16, \$16	\$8, \$12
Large	\$18, \$9	\$12, \$8	\$0, \$0

N.E at (Small, Small).

Q: What if Alpha moved first?

The game tree, and first-mover advantage.

If Alpha preempts Beta, then use the *game tree*:

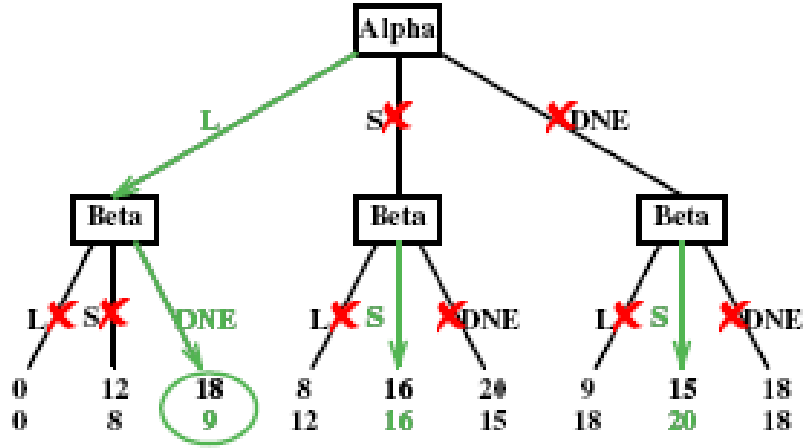


Figure 1. Game Tree, Payoffs: Alpha's, Beta's

SPE

. at Alpha: Large; Beta: Don't Expand, (not S,S).
Payoffs now ($18, $9) instead of ($16, $16):
Alpha is $2 better off, but Beta is $7 worse off.

Order Advantages

In the Capacity Game, Alpha gains $2 by preempting Beta: *first-mover advantage*.

In some interactions, however, there is *second-mover advantage*: a catalogue company whose catalogue came out later could undercut its rival whose catalogue and prices were announced first.

First-mover advantage comes from the ability to *commit* to an advantageous position and to force other players to adapt to it.

Second-mover advantage comes from the *flexibility* to adapt oneself to the others' choices.

Commitment v. flexibility?

Evidence of Rollback

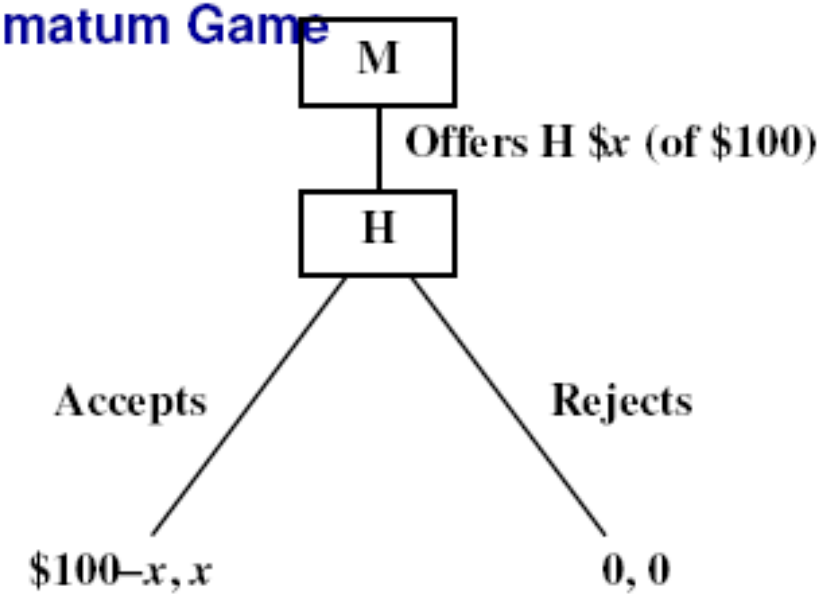
The Ultimatum Game:

Players and game: Mortimer and Hotspur are to divide \$100 between themselves. The game structure is common knowledge.

- Mortimer offers Hotspur an amount \$$x$ of the \$100. Then —
- *either* Hotspur accepts \$$x$, and Mortimer receives the remainder of the \$100, and the game ends;
- *or* Hotspur rejects \$$x$, and neither gets anything.

What will Mortimer offer \$$x$?
What would you offer? (Write it down.)

The Ultimatum Game



Most offer a 50:50 split, and almost all accept. Most reject less than $25 offered, and some even $40.

A fairness (equal) focal point.

Voting Revisited

Gore v. Bush
before the U.S. Supreme Court

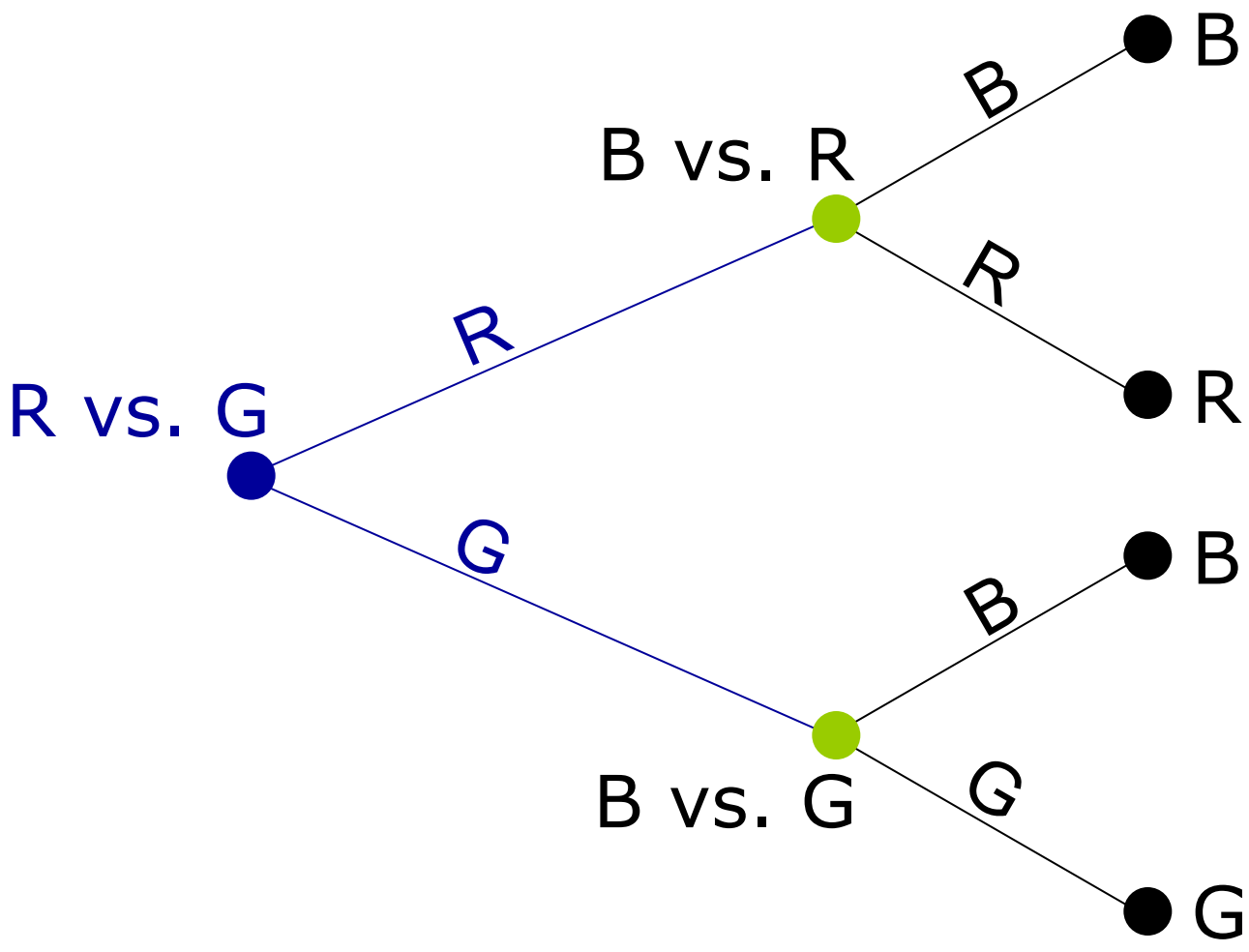
- Recall member preferences:

  4 (B>G>R) 3 (G>R>B) 2 (R>B>G)

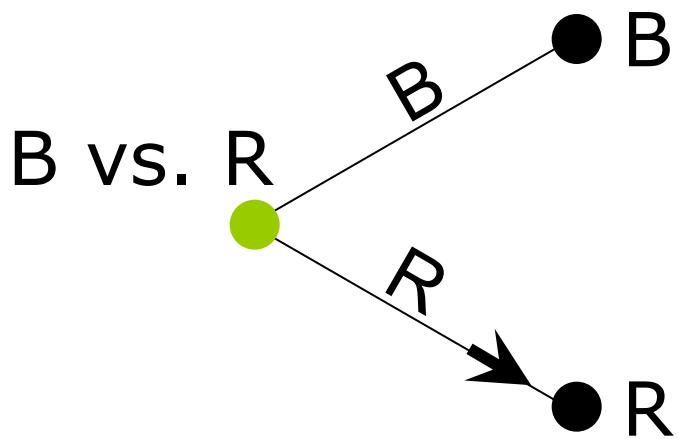
- Majority rule results:
  - B beats G ; G beats R ; R beats B

- Voting results (example):
  - R beats G then winner versus B → B

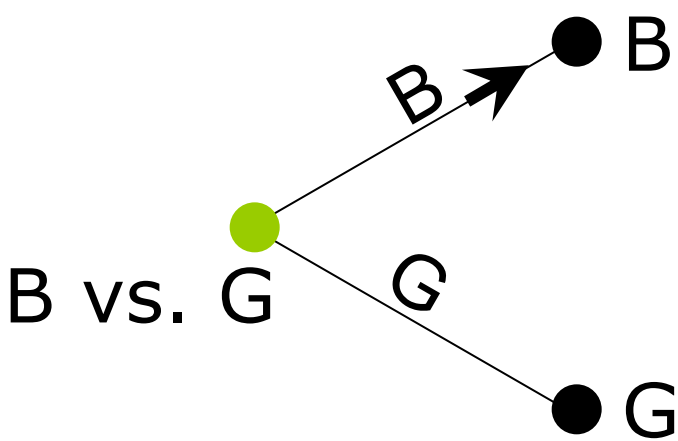
Voting as a Sequential Game



Looking Forward …



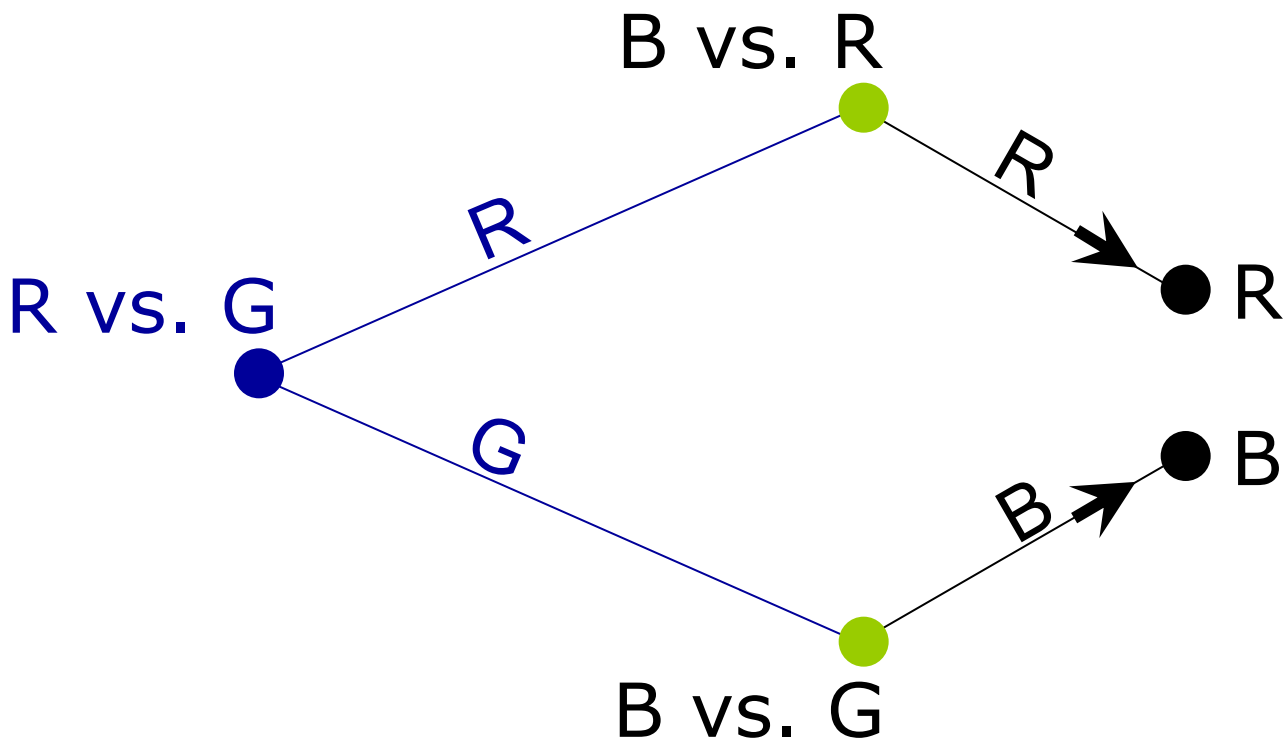
A majority prefers R to B



A majority prefers B to G

… And Reasoning Back

Three judges prefer G to R to B.
How should they vote in the first round?



What Happened?

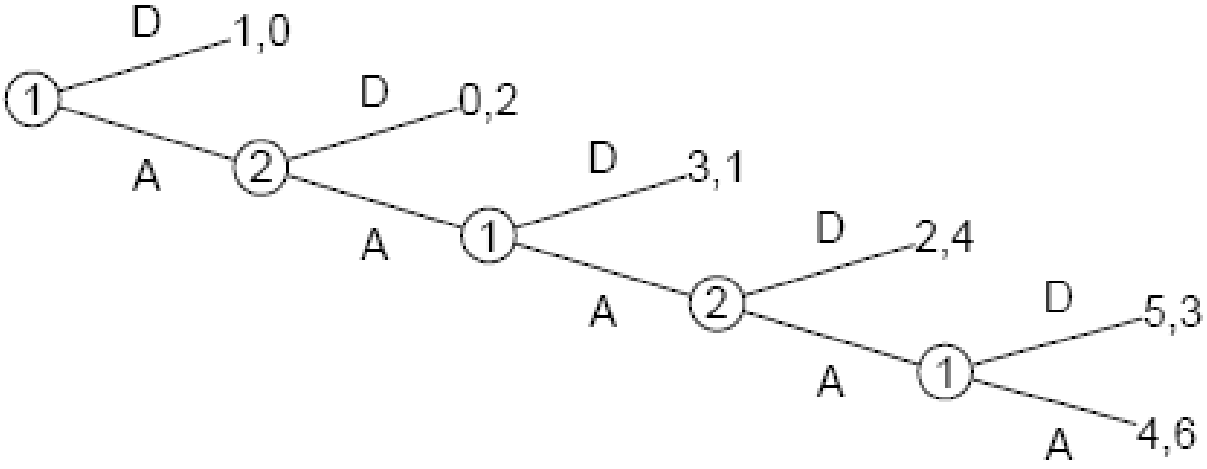
- Gore supporters have preferences
  - **Gore > Remand > Bush**
- First round vote between Remand and Gore
  - Gore supporters prefer Gore
  - But a vote for Gore is *in effect* a vote for Bush!
- So Gore supporters vote for remand.
  - Guarantee themselves second best choice
-

Examples of Strategic Voting

Not necessarily good to vote your true preferences

- Outcome is still predetermined:
  - B beats G then winner versus R → ~~R~~ G
  - B beats R then winner versus G → ~~G~~ B
  - G beats R then winner versus B → ~~B~~ R

Centipede Game



Sequential Rationality

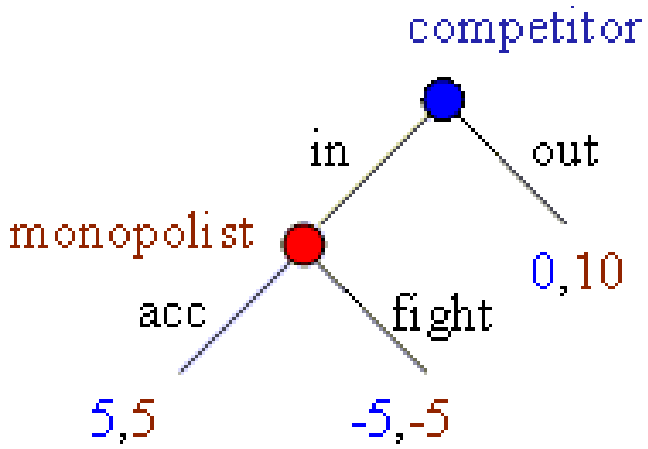
COMMANDMENT

Look forward and reason back.

Anticipate what your rivals will do
tomorrow
in response to your actions
today

Question 1.

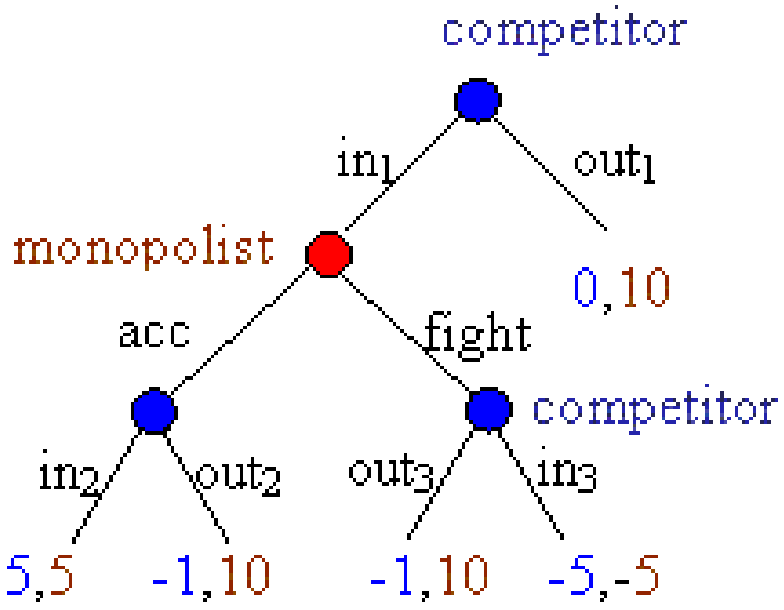
A firm is quite happy monopolizing its industry with profits of $10M. A potential competitor considers entering the industry. If the competitor elects not to enter, it earns profits of $0 and the monopolist maintains its profit of $10M. If the competitor enters, the monopolist must either accommodate the entry or fight. If the monopolist accommodates, both firms earn $5M. If the monopolist fights, both firms lose $5M. The game is represented by the fo



What is the rollback equilibrium of the above game?

- ✔ {In,Accommodate}
- {In,Fight}
- {Out,Accommodate}
- {Out,Fight}

Consider the above game (question 1) but suppose that the decision to enter by the competitor is reversible in the following sense: after it has entered, and after the monopolist has chosen to accommodate or fight, the competitor can choose to remain in the industry (and receive either the $5M profits or $5M loss) or to exit. Suppose that exiting at this point results in a loss to the entrant of $1M, and the monopolist regains its $10M profit. The new game is represented by the following tree:



Question 4.

The three person executive committee of the Owen Honor Council is considering a case of cheating on a take home exam. The members decide that, for the sake of consistency, they should first determine if such an offense warrants a strong or a weak punishment, and then determine if the accused should receive that punishment. Hence, the order of the voting will be:

- Round 1. Strong versus weak sanctions
- Round 2. Winner of round 1 versus no sanction

The three members vote by majority rule.

If the judges vote Myopically (i.e. each votes for his or her true favorite option in every round), which alternative will prevail?

The three members of the committee have the following preferences:

Person 1 prefers:	strong	to	no sanction	to	weak
Person 2 prefers:	no sanction	to	weak	to	strong
Person 3 prefers:	weak	to	strong	to	no sanction

No sanction

✔ Weak

Strong

It depends

Question 5.

The three judges from the previous question decide that prior to voting, they will consult a game theorist to determine how to vote strategically. If, rather than voting myopically, the judges look forward and reason back, voting strategically in the first round, which outcome will prevail?

No sanction

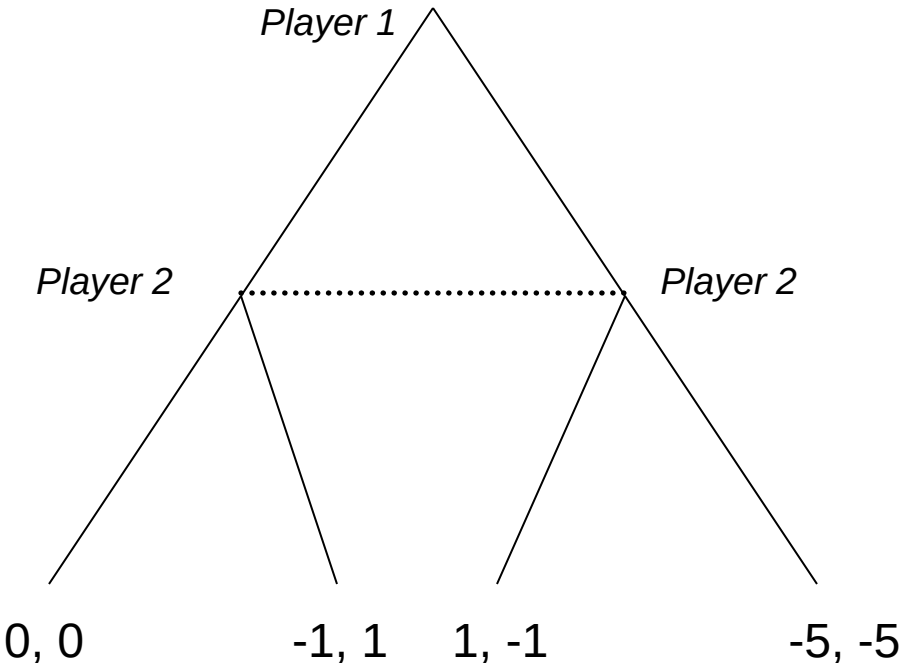
Weak

✔ **Strong**

It depends

Imperfect information

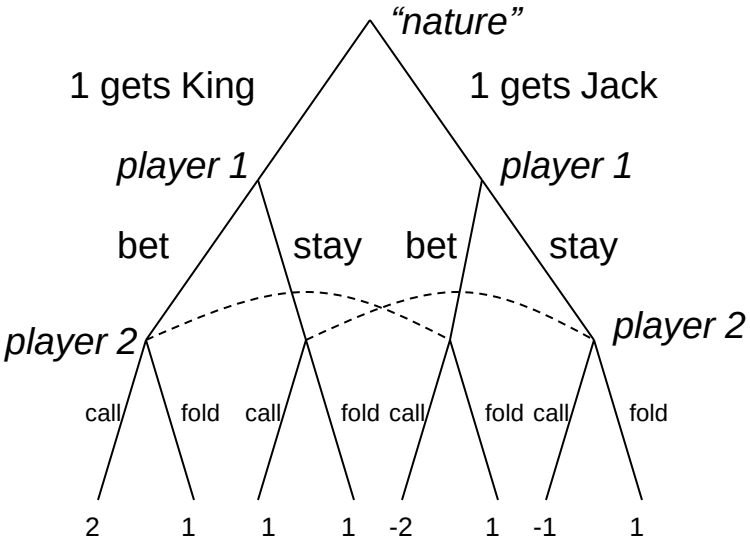
- Dotted lines indicate that a player cannot distinguish between two (or more) states
  - A set of states that are connected by dotted lines is called an information set
- Reflected in the normal-form representation



	L	R
L	0, 0	-1, 1
R	1, -1	-5, -5

- Any normal-form game can be transformed into an imperfect-information extensive-form game this way

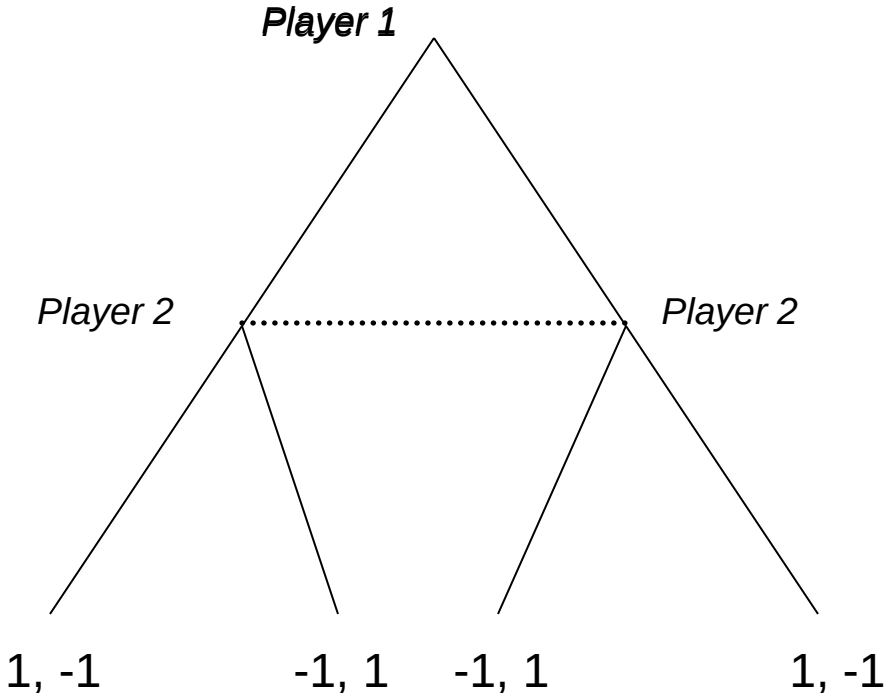
A poker-like game



		2/3 cc	cf	1/3 fc	ff
1/3	bb	0, 0	0, 0	1, -1	1, -1
2/3	bs	.5, -.5	1.5, -1.5	0, 0	1, -1
	sb	-.5, .5	-.5, .5	1, -1	1, -1
	ss	0, 0	1, -1	0, 0	1, -1

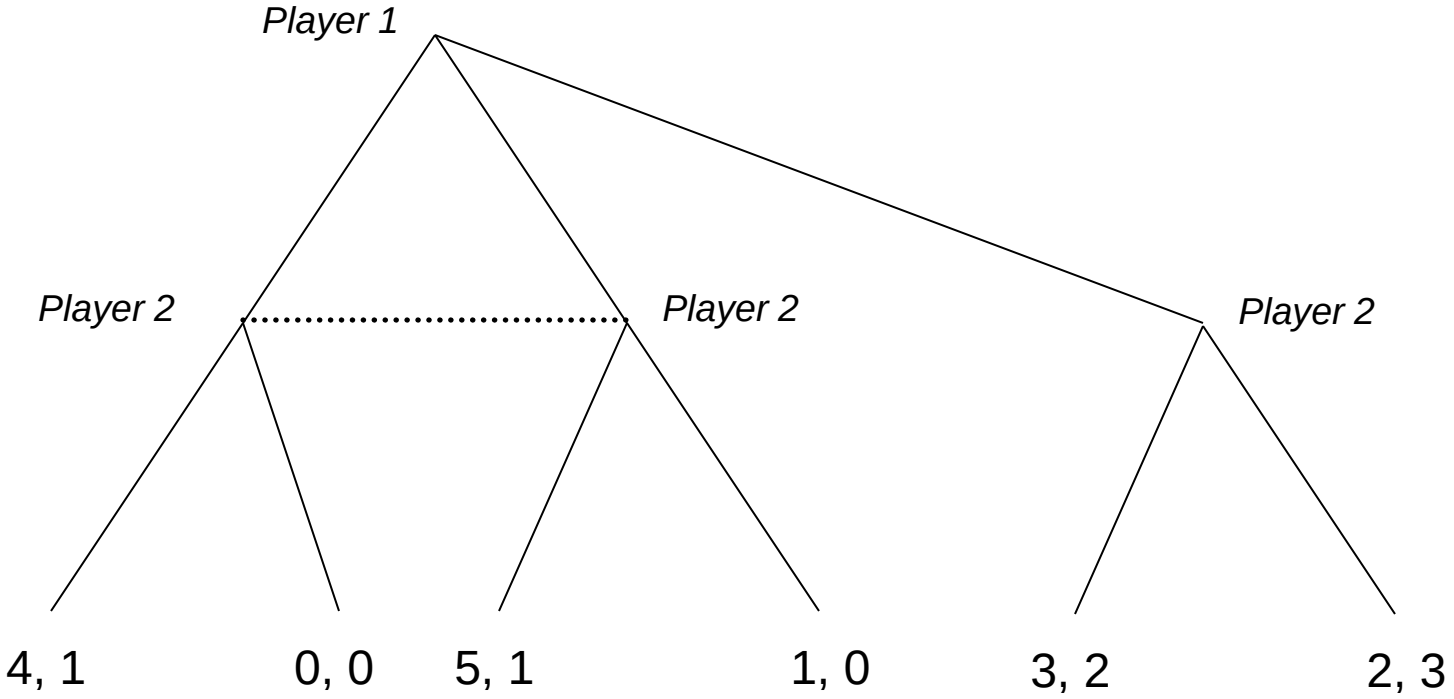
Subgame perfection and imperfect information

- How should we extend the notion of subgame perfection to games of imperfect information?



- We cannot expect Player 2 to play Right after Player 1 plays Left, and Left after Player 1 plays Right, because of the information set
- Let us say that a subtree is a subgame only if there are no information sets that connect the subtree to parts outside the subtree

Subgame perfection and imperfect information…



- One of the Nash equilibria is: (R, RR)
- Also subgame perfect (the only subgames are the whole game, and the subgame after Player 1 moves Right)
- But it is not reasonable to believe that Player 2 will move Right after Player 1 moves Left/Middle (not a credible threat)
- There exist more sophisticated refinements of Nash equilibrium that rule out such behavior

Computing equilibria in the extensive form

- Can just use normal-form representation
  - Misses issues of subgame perfection, etc.
- Another problem: there are exponentially many pure strategies, so normal form is exponentially larger
  - Even given polynomial-time algorithms for normal form, time would still be exponential in the size of the extensive form
- There are other techniques that reason directly over the extensive form and scale much better
  - E.g., using the sequence form of the game

Commitment

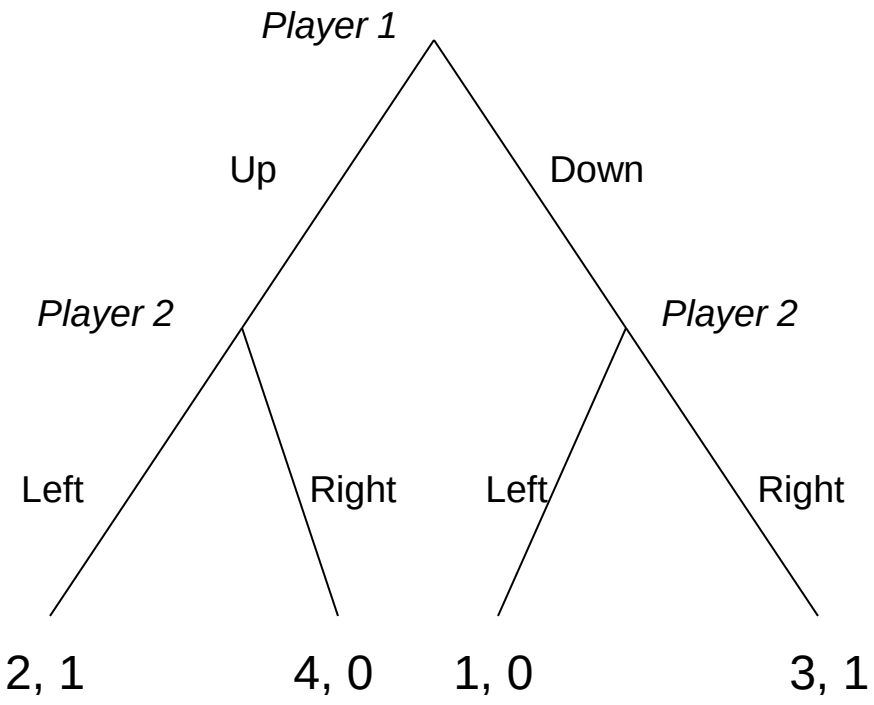
- Consider the following (normal-form) game:

2, 1	4, 0
1, 0	3, 1

- How should this game be played?
- Now suppose the game is played as follows:
  - Player 1 commits to playing one of the rows,
  - Player 2 observes the commitment and then chooses a column
- What is the optimal strategy for player 1?
- What if 1 can commit to a mixed strategy?

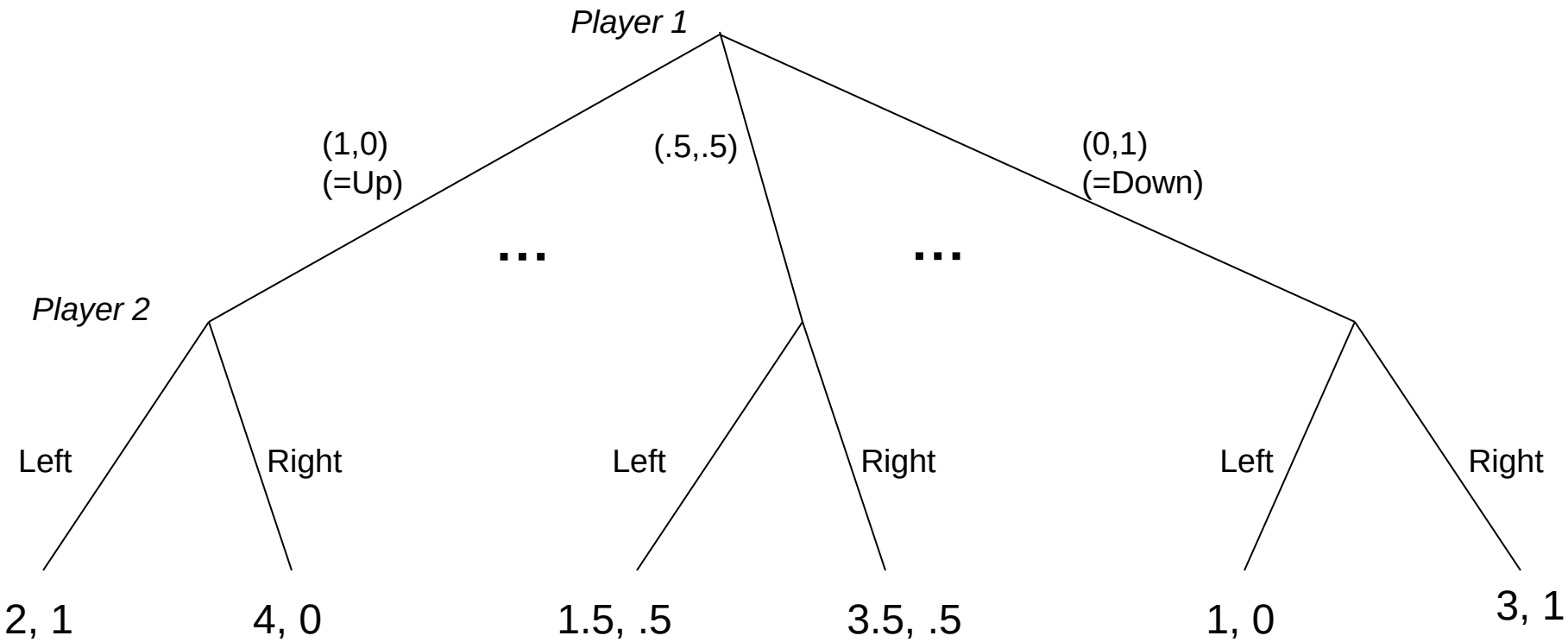
Commitment as an extensive-form game

- For the case of committing to a pure strategy:



Commitment as an extensive-form game

- For the case of committing to a mixed strategy:



- Infinite-size game; computationally impractical to reason with the extensive form here

Solving for the optimal mixed strategy to commit to

[Conitzer & Sandholm 2006; see also: von Stengel & Zamir 2004, Letchford, Conitzer, Munagala 2009]

- For every column t separately, we will solve separately for the best mixed row strategy (defined by $\mathbf{p_s}$) that induces player 2 to play t
- maximize $\Sigma_s \mathbf{p_s} u_1(s, t)$
- subject to

  for any t', $\Sigma_s \mathbf{p_s} u_2(s, t) \geq \Sigma_s \mathbf{p_s} u_2(s, t')$

  $\Sigma_s \mathbf{p_s} = 1$
- (May be infeasible, e.g., if t is strictly dominated)
- Pick the t that is best for player 1